

Presenter Notes

Online PAC Labeling from a Linear Programming Perspective

Companion to `slides.tex` — target ≈ 15 minutes

Pacing guide. Setup + LP (0:00–4:00), offline + results (4:00–11:30), then the *online segment* (Slides 12–17) as a self-contained ≈ 5 -minute block. Parenthetical times are *cumulative* through Slide 11; for the online block they restart and are *relative to the start of the segment* (it is a separate speaker). If running long, compress the dual derivation (Slide 5) and the proof sketches (Slides 9 and 15) and lean on the figures — those are the parts an audience can skip without losing the thread.

The one-sentence thesis (keep returning to it): *a single LP dual variable λ — the “price of error” — unifies offline certification and online auditing, and proxies only ever affect cost, never the error guarantee.*

Slide 1. Title

($\approx 0:20$) **Open simply.** “Our project revisits PAC labeling — cheaply labeling a large dataset with a finite-sample error guarantee — but through the lens of a linear program. That single change of viewpoint buys us multi-source routing and an online extension almost for free.” Name the three authors, then move on.

Backup — “PAC” = probably approximately correct: the error guarantee holds with probability $1 - \alpha$ over the calibration (offline) or audit (online) randomness. Our starting point is PAC labeling of Candès et al. (2025); our contribution is the optimization-first framing, item/source-specific costs, and the audited online extension.

Slide 2. The problem: labeling a big pool, cheaply

($\approx 1:30$) **Set the stakes.** We have a fixed pool of T unlabeled items $X_1, \dots, X_T$ with unknown true labels $Y_1, \dots, Y_T$, and we must release a label for *every* item. Two ways to get a label: k cheap ML models (free but fallible) or a human **expert** (treated as ground truth, but expensive). We want to lean on the AI as much as possible while *certifying* that the average error of the released dataset is below a user-chosen target ε .

State the tension — it organizes the whole talk. More AI lowers cost; more expert lowers error. Everything that follows is about navigating that frontier *with a guarantee*, not just a heuristic. The headline numbers to foreshadow: at a 10% error target we skip the expert on roughly 78% of ImageNet items.

Why “LP”? Because the trade-off is literally a constrained cost-minimization: minimize expert queries subject to an error budget. Writing it as an LP and taking the dual is what collapses a $T \times (k+1)$ decision problem down to tuning one scalar.

Slide 3. Setup and notation

($\approx 3:00$) **Walk the two boxes.** Action set $\mathcal{A} = \{E, 1, \dots, k\}$: defer to the expert E , or accept

one of k cheap models that produce predictions $\hat{Y}_t^{(j)}$. Two item-wise primitives:

$$h_t(a) = \begin{cases} 0, & a = E \\ \ell(Y_t, \hat{Y}_t^{(a)}), & a \in [k] \end{cases} \quad c_t(a) = \begin{cases} 1, & a = E \\ 0, & a \in [k]. \end{cases}$$

Emphasize the asymmetry — it drives every later simplification. The expert has zero loss but unit cost; cheap models have zero cost but possibly positive loss. The labeling loss ℓ is bounded and normalized to $[0, 1]$. The released label is $\tilde{Y}_t = Y_t$ when we defer, else $\hat{Y}_t^{(a_t)}$.

Backup — The unit/zero cost split is the “equal cheap-source cost” regime ($c_t(j) = 0$ for all j , $c_t(E) = 1$). The *LP itself* permits item- and source-specific costs; we specialize to equal costs because that is what makes $L_T(\lambda)$ monotone and what the data supports. Unequal costs come back in the limitation slide and in the Q&A.

Slide 4. The cost-aware LP

($\approx 4:00$) **Read the LP in words, not symbols.** Decision variable $x_{t,a} \in [0, 1]$ is the probability of taking action a on item t . We *minimize average cost* $\frac{1}{T} \sum_{t,a} c_t(a)x_{t,a}$ subject to: (i) the *average error budget* $\frac{1}{T} \sum_{t,a} h_t(a)x_{t,a} \leq \varepsilon$; (ii) each item’s probabilities sum to one; (iii) nonnegativity.

Two things to stress. First: there is exactly *one* coupling constraint — the global error budget — and that single coupling is what makes the dual one-dimensional. Everything else is per-item (a simplex). Second: this is the *oracle* LP — if the true losses $h_t(a)$ were known, its solution *is* the optimal labeling policy. The rest of the talk is about not knowing $h_t(a)$ and plugging in proxies $\hat{h}_t(a)$.

Backup — The LP is a continuous relaxation, but its optimum is attained at a vertex, so the per-item solution is essentially deterministic (an item is routed, deferred, or sits exactly at a threshold). This is why the eventual rule is a hard threshold, not a randomized mixture.

Slide 5. The dual / price of error

($\approx 5:00$) **This is the conceptual heart — slow down here.** Put a single multiplier $\lambda \geq 0$ on the error constraint. The reduced dual is the one-dimensional *concave* maximization

$$\max_{\lambda \geq 0} D(\lambda) := -\lambda\varepsilon + \frac{1}{T} \sum_{t=1}^T \min_{a \in \mathcal{A}} \{c_t(a) + \lambda h_t(a)\}.$$

The structural payoff: *for fixed λ the inner minimization separates across items*, giving the fixed-price rule $a_t(\lambda) \in \arg \min_a \{c_t(a) + \lambda h_t(a)\}$.

Interpretation to say out loud. λ is a shadow price — the exchange rate between a unit of cost and a unit of error. Push λ up and error becomes “expensive,” so we defer more; push it down and we trust the AI. A high-dimensional policy collapses to tuning one scalar.

Backup — Dual derivation if pressed: introduce $\lambda \geq 0$ for the error constraint and a *free* multiplier $\mu_t \in \mathbb{R}$ for each simplex equality; keep $x \geq 0$ as the domain. Linearity in each $x_{t,a}$ forces the dual feasibility condition $\mu_t \leq \frac{1}{T} \{c_t(a) + \lambda h_t(a)\}$ for all a (else the inner inf is $-\infty$). The objective increases in each μ_t , so optimally $\mu_t^*(\lambda) = \frac{1}{T} \min_a \{c_t(a) + \lambda h_t(a)\}$, which substitutes to give $D(\lambda)$. Strong duality holds: the primal is feasible (defer everything $\Rightarrow$ zero error) and bounded (costs ≥ 0), so finite-dimensional LP duality applies. We assume losses in the argmin are distinct so ties don’t occur.

Slide 6. Threshold form

($\approx 6:00$) **Make the rule concrete — this is the rule we actually run.** Under equal cheap costs the chosen cheap model doesn't depend on λ , so the rule factors into two steps. **Step 1, route:** pick the best cheap model $j_t^* \in \arg \min_j \hat{h}_t(j)$ and read off its score $s_t = \min_j \hat{h}_t(j)$ — think of s_t as an “uncertainty.” **Step 2, threshold:** λ only sets a cutoff $1/\lambda$:

$$s_t \geq 1/\lambda \Rightarrow \text{defer to expert}, \quad s_t < 1/\lambda \Rightarrow \text{accept the routed cheap label.}$$

The payoff line. With a single source and $\hat{h}_t = \text{uncertainty}$, this is *exactly* the original PAC-labeling threshold family. Multiple sources cost nothing extra — the score just becomes $\min_j \hat{h}_t(j)$. And the very same scalar will be *fixed once* offline or *moved over time* online — that's the fork in the talk.

Backup — We replace the unknown true loss by a proxy $\hat{h}_t(a)$ — an uncertainty score, a calibrated error estimate, anything *predictable* from past audits and current non-expert info. Crucially proxies need *not* be calibrated or conservative: a better proxy improves cost, while held-out calibration provides validity. In the experiments $\hat{h} = 1 - \text{confidence}$.

Slide 7. Part I divider — Offline

($\approx 6:15$) **One sentence.** “In the offline case the whole pool is in front of us: we spend a handful of expert labels to *certify* one price, then label everything at that fixed price.”

Slide 8. Offline calibration

($\approx 8:00$) **The split that makes it work.** Decompose the price- λ policy into its realized loss $H_t(\lambda) = (1 - D_t^\lambda) \ell(Y_t, \hat{Y}_t^{(j_t^*)})$ and cost $C_t(\lambda) = D_t^\lambda$. The average *cost* $C_T(\lambda)$ is fully known from proxies and costs — no labels needed. Only the average *error* $L_T(\lambda) = \frac{1}{T} \sum_t H_t(\lambda)$ needs true labels, so that is the one thing we estimate. That is precisely what the calibration set buys.

Three steps. (1) Sample m calibration indices i.i.d. uniformly from the pool (with replacement), pay for their expert labels, and form the empirical loss $\hat{L}_m(\lambda) = \frac{1}{m} \sum_s H_{I_s}(\lambda)$. (2) Add a Hoeffding slack for a one-sided upper confidence bound $U_m(\lambda; \alpha) = \hat{L}_m(\lambda) + \sqrt{\log(1/\alpha)/2m}$. (3) Pick the *cheapest certified price* $\hat{\lambda} = \min\{\lambda \geq 0 : U_m(\lambda; \alpha) \leq \varepsilon\}$, falling back to expert-only ($\lambda = \infty$) if none qualifies.

Why we pay no union-bound tax — this is the slick part. As λ increases, the threshold $1/\lambda$ *decreases*, so each item can only move from AI to expert, never between two cheap models. Hence $H_t(\lambda)$ is non-increasing and $C_t(\lambda)$ non-decreasing in λ : the policies form one *nested, monotone* family. A single pointwise Hoeffding bound, placed at the relevant crossing price, certifies the entire continuum — no grid, no $\log |\Lambda|$ penalty. Because U_m is non-increasing and C_T non-decreasing, the “cheapest certified price” is also the cost-minimizer.

Backup — Implementation: $m = 0.2T$, $\alpha = 0.05$, $\varepsilon \in \{0.05, 0.10, 0.15\}$. The sweep uses a grid (expert-only plus 201 evenly spaced thresholds) purely for plotting — the monotone certificate needs no grid. The original-PAC baseline uses the same calibration sample and the same radius $\sqrt{\log(1/\alpha)/2m}$ but picks its threshold from the calibration mistake ordering rather than a grid.

Slide 9. Offline guarantee

($\approx 9:00$) **State the theorem plainly.**

With probability $\geq 1 - \alpha$ over the calibration draw, the realized average error on the *whole* pool satisfies $\frac{1}{T} \sum_t \ell(Y_t, \tilde{Y}_t) \leq \varepsilon$. Finite-sample, distribution-free, and it safely falls back to expert-only when no finite price qualifies.

Proof intuition in one breath. Order the finitely many distinct policies by λ . Let λ° be the last (fixed, data-independent) policy whose *true* loss still exceeds ε . A bad selection $L_T(\hat{\lambda}) > \varepsilon$ forces $\hat{\lambda}$ to land no later than λ° ; since U_m is non-increasing, $U_m(\lambda^\circ) \leq U_m(\hat{\lambda}) \leq \varepsilon < L_T(\lambda^\circ)$. So the bad event is contained in a *single* fixed-price Hoeffding failure $\{L_T(\lambda^\circ) > U_m(\lambda^\circ)\}$, which has probability $\leq \alpha$. Monotonicity is doing all the work — that’s why no union bound.

Generalization corollary — mention briefly. If calibration is i.i.d. from a distribution P , the *population* risk at $\hat{\lambda}$ is $\leq \varepsilon$ w.p. $1 - \alpha$; and on a fresh deployment pool of size n the realized error is $\leq \varepsilon + \sqrt{\log(1/\delta)/2n}$ w.p. $1 - \alpha - \delta$. **Repeat the slogan:** proxies move cost, not validity — a worse proxy just defers more often.

Slide 10. Offline experiment 1 — one source

($\approx 10:30$) **Setup in a breath.** Two ImageNet prediction tables —

ImageNet ($T = 50,000$) and ImageNetV2 ($T = 10,000$) — each with a ground-truth label, a model prediction, and a confidence. Cheap source = the model prediction, expert = ground truth, proxy loss $\hat{h} = 1 - \text{confidence}$. Every number is an average over **500 random trials**; the slide shows $\varepsilon = 0.10$.

Read the table left to right. LP fixed-price vs. original PAC: error 0.0873 vs. 0.0879 on ImageNet, budget saved 0.7848 vs. 0.7864 — essentially identical, with the *violation column exactly zero*. That is the sanity check: with one source the LP rule *is* PAC, exactly as Slide 6 predicts. We skip the expert on $\sim 78\%$ of ImageNet items, $\sim 55\%$ on the harder ImageNetV2.

Point at the figure. Across $\varepsilon \in \{0.05, 0.10, 0.15\}$ the two methods’ budget-saved and realized-error curves sit on top of each other; realized error hugs — and stays under — the gray target line. Higher ε buys more savings ($0.63 \rightarrow 0.79 \rightarrow 0.89$ on ImageNet).

Slide 11. Offline experiment 2 — two sources

($\approx 11:30$) **This is the slide that justifies the LP framing — give it weight.** The real data has only one genuine source, so we *synthesize* two complementary ones: partition items by the parity of the true class, and let each source reuse the original predictor on its specialty half while emitting a random low-confidence label off it. The pool is fixed across trials. Now per-item routing genuinely matters.

The contrast — the money numbers. At $\varepsilon = 0.10$ on ImageNet, LP routing saves **0.7848** of the expert budget versus **0.4746** for the best single source — and that baseline is already given a fair shake via a Bonferroni correction (run both singles at $\alpha/2$, keep the better). Same error control (violation = 0), but the LP picks the right source per item and defers far less. The pattern holds across ε and on ImageNetV2 (0.5457 vs. 0.3587 at $\varepsilon = 0.10$).

Takeaway line. “A scalar threshold on a single source leaves a lot of budget on the table; the LP recovers it by routing first, then thresholding.”

Backup — The monotone certificate still applies because, under equal costs, the routed source $j_t^* = \arg \min_j \hat{h}_t(j)$ doesn’t depend on λ — the score is just $\min_j \hat{h}_t(j)$ and the policy is still nested in λ . The

grid here is 201 thresholds on $\min_j \hat{h}_t(j)$ plus expert-only; same $m = 0.2T$, $\alpha = 0.05$.

Slide 12. Part II divider — Online

($\approx 0:10$ (segment-relative)) **One sentence.** “Now items stream in one at a time, and an AI label’s true error is invisible unless we pay to audit it — so we can no longer certify a fixed price up front; the price has to *move*.” This begins your ≈ 5 -minute segment; the times below are relative to here.

Slide 13. Online setup: moving threshold + audits

($\approx 0:55$) **The obstacle first.** The new difficulty versus offline: once we release an AI label we normally *never learn* whether it was right — the loss $\ell(Y_t, \hat{Y}_t)$ is hidden unless we pay the expert. So we cannot just read off our running error rate; we have to buy information.

Describe the loop concretely. Each round hold a current price λ_t , i.e. a threshold $1/\lambda_t$ on the routed uncertainty score $s_t = \min_j \hat{h}_t(j)$. Route to the lowest-proxy model; if it is too uncertain ($s_t \geq 1/\lambda_t$) defer to the expert (always correct). Otherwise release the AI label, then flip a coin: with probability p *audit* it — pay the expert, *correct* the released label, and *observe the AI’s true loss* ℓ_t . Audits are rare on purpose, to stay cheap; that observed ℓ_t is the only feedback, and it drives the next slide’s update.

Backup — Predictability (Assumption 1): routing, the audit probability, and the selected source use only past information plus the current non-expert info $(X_t, \{\hat{Y}_t^{(j)}, \hat{h}_t(j)\})$; given that, the audit draw $A_t \sim \text{Bernoulli}(p_t)$ is independent, with $p_t \in [p_{\min}, 1]$ (in the experiments $p_t \equiv p$ is constant). The proxy is also floored, $\hat{h}_t(j) \in [\rho, 1]$ — that floor is what bounds the price on Slide 15.

Slide 14. The price update

($\approx 2:10$) **The update is one-line dual ascent.** $\lambda_{t+1} = [\lambda_t + \eta \hat{Z}_t]_+$, projected to stay ≥ 0 , where $\hat{Z}_t$ is this round’s feedback. Three cases: $-\varepsilon$ on a deferral; $(\ell_t - \varepsilon)/p$ on an audited AI label; 0 on an unaudited AI label.

Narrate the sign. An audited loss *above* target ($\ell_t > \varepsilon$) makes $\hat{Z}_t > 0$, so λ rises, the threshold $1/\lambda$ falls, and future routing gets more cautious; a deferral or a clean audit ($\ell_t < \varepsilon$) lowers λ and leans on the AI more.

Why the $1/p$ — the key idea. Audits are rare, so each audited item must stand in for the $\sim 1/p$ unaudited ones: inverse-probability weighting. That makes the update *unbiased* — in expectation it behaves as if every AI loss were seen, $\mathbb{E}[\hat{Z}_t \mid \text{past}] = (\text{AI error}) - \varepsilon$. **Flag the simplification:** the update now centers *directly* on the final target ε ; there is no separate internal target ε_0 any more (the report dropped it). The only gap to ε is the vanishing slack on the next slide.

Backup — In the proof’s notation the three cases compress to the centered inverse-observation signal $\hat{Z}_t = \frac{O_t}{q_t}(H_t - \varepsilon)$, with mask $O_t = D_t + (1 - D_t)A_t$ and $q_t = D_t + (1 - D_t)p_t$ ($q_t = 1$ on a deferral, p_t on an AI decision). Proposition 1: $\mathbb{E}[O_t/q_t \mid \mathcal{G}_t] = 1$ in both cases ($1/1$ and p_t/p_t), hence $\mathbb{E}[\hat{Z}_t \mid \mathcal{G}_t] = H_t - \varepsilon$. Keep $O_t, q_t, \mathcal{G}_t$ off the slides — they are bookkeeping for the martingale step only.

Slide 15. Finite-horizon guarantee: anatomy of the bound

($\approx 3:35$) **State the result.** For *any* data sequence — no distributional assumptions, the only

randomness is the audit coins — with probability $\geq 1 - \delta$,

$$\frac{1}{T} \sum_t \ell(Y_t, \tilde{Y}_t) \leq \varepsilon + B_T, \quad B_T = \frac{\Lambda_{\text{on}}}{\eta T} + \frac{1}{p_{\min}} \sqrt{\frac{2 \log(1/\delta)}{T}}.$$

It controls *realized* error, not regret against an offline oracle.

Read the bound term by term — this is the slide’s whole job. (1) ε is the target the update steers toward. (2) $\Lambda_{\text{on}}/(\eta T)$ is optimization error — how far the price can wander before it settles — decaying like $1/T$. (3) the last term is the *price of partial feedback*: sampling noise from auditing only a fraction, scaling like $1/\sqrt{T}$ and inflating as the audit floor $p_{\min}$ shrinks.

Define Λ_{on} and give the intuition (the new piece on the slide). Explicitly $\Lambda_{\text{on}} = \max\{\lambda_1, 1/\rho + \eta(1 - \varepsilon)/p_{\min}\}$ — a constant ceiling the price can never exceed. *Why* it can’t run away: the proxy is floored at ρ , so the moment λ_t reaches $1/\rho$ the threshold $1/\lambda_t$ drops to ρ or below; then *every* item looks too uncertain and defers, which only pushes the price back down. A single update can overshoot that level by at most one step $\eta(1 - \varepsilon)/p_{\min}$ — hence the formula. It is self-limiting.

The slack vanishes. $B_T = O(T^{-1/2})$, so the released error approaches ε as the stream grows; with $\eta \asymp T^{-1/2}$ that rate is explicit. **Released vs. AI error:** the bound is really on the counterfactual AI error, but every audited mistake is corrected before release, so the shipped error is only smaller.

Backup — Proof sketch (only if asked): (i) the price is bounded, $0 \leq \lambda_t \leq \Lambda_{\text{on}}$, by the deferral argument above. (ii) Since $[x]_+ \geq x$, telescoping gives $\sum_t \hat{Z}_t \leq \Lambda_{\text{on}}/\eta$. (iii) The gap $M_t = (H_t - \varepsilon) - \hat{Z}_t$ is a martingale-difference sequence (Prop. 1) bounded by $1/p_{\min}$, so Azuma–Hoeffding gives $\sum_t M_t \leq \frac{1}{p_{\min}} \sqrt{2T \log(1/\delta)}$. (iv) Combine via $\sum_t H_t = T\varepsilon + \sum_t \hat{Z}_t + \sum_t M_t$, then use audit-and-correct’s pointwise $G_t \leq H_t$ to pass from AI error to released error.

Slide 16. Online experiment

($\approx 4:20$) **One clean message: p is a validity–cost knob.** Per trial we randomly permute the pool into a stream; settings $\eta = 1$, proxy floor $\rho = 0.01$, $\lambda_1 = 1/\varepsilon$, audit rate $p \in \{0.1, 0.2, 0.5\}$, again 500 trials. At $\varepsilon = 0.10$ on ImageNet: $p = 0.1$ saves 0.6629 at final error 0.0780; $p = 0.2$ saves 0.6115 at 0.0737; $p = 0.5$ saves only 0.3952 but drives final error down to 0.0490. Smaller p = cheaper; larger p = more corrections, lower released error.

Point at the dashed vs. solid curves. Dashed = counterfactual AI error *before* correction (hovers near ε , this is the H_t the theory bounds); solid = realized error *after* audit-and-correct (strictly below). The gap between them is literally the value of auditing. The released-label error stays controlled even under heavily partial feedback — the online guarantee in action.

Backup — Because the update centers on ε directly (no ε_0 slack is subtracted), the bound sits at exactly $\varepsilon + B_T$, so at small ε and small p finite- T fluctuation can produce a tiny violation (e.g. ImageNetV2, $\varepsilon = 0.05$, $p = 0.1$: violation 0.184); it shrinks as T grows or p rises. The “AI error” column is the pre-correction counterfactual; “final error” is post-correction.

Slide 17. Takeaways

($\approx 4:45$) **Close on the unifying idea.** One LP dual gives a single price of error. Offline we *certify* it on held-out labels (Hoeffding + monotonicity); online we *move* it with audited inverse-probability feedback (dual ascent + Azuma). In both regimes proxies affect efficiency, never validity.

The evidence in three beats. LP matches PAC with one source; routing clearly wins with two (0.78 vs. 0.47 budget saved); the online rule stays valid under partial feedback with p as the cost/error dial.

Honest limitation — say it plainly. Our real data has only one genuine cheap source, so the two-source win uses a synthetic split. The genuinely interesting regime is *many sources with heterogeneous costs*: there the routed source $j_t^\lambda = \arg \min_j \{c_t(j) + \lambda \hat{h}_t(j)\}$ depends on λ , source-switching can make $L_T(\lambda)$ non-monotone, and the clean single-point certificate breaks — you fall back to certifying a finite price grid (or enumerating breakpoints). Then thank the audience and invite questions.

Anticipated Questions & Prepared Answers

Q1. Why bother with the LP at all if one source just recovers original PAC?

Because the LP is what *tells* you the right multi-source rule and where the guarantee comes from. With one source it must coincide with PAC — that’s the sanity check, not the contribution. The value shows up the moment you have more than one source (or non-trivial costs): the dual hands you “route to $\arg \min_a \{c_t(a) + \lambda \hat{h}_t(a)\}$, then threshold” automatically, and the two-source experiment shows that buys ~ 0.31 extra budget saved over the best single source at the same error.

Q2. The two cheap sources are synthetic — is the gain real?

Fair, and we flag it as the main limitation. The construction (specialty by class parity, random off-specialty labels) is engineered so routing *can* help, so the number is an illustration of the mechanism, not a benchmark claim. But the mechanism is real: whenever sources have complementary strengths, picking the lower-proxy source per item before deferring strictly dominates committing to one source. The honest next step is heterogeneous-cost data with several genuine models.

Q3. The expert is assumed perfect. What if experts are noisy?

We treat the expert label as ground truth ($h_t(E) = 0$) — that’s the standard PAC-labeling abstraction. With noisy experts the “error” being certified is error *relative to the expert oracle*, and you’d need to model expert disagreement (e.g. multiple experts, majority vote, or a known expert error rate) to recover an absolute guarantee. Nothing in the LP forbids $h_t(E) > 0$; the clean fallback “defer everything $\Rightarrow$ zero error” is what would need revisiting.

Q4. Hoeffding is loose. Can you tighten the offline bound?

Yes — any valid one-sided upper confidence bound slots into the same monotone argument. We note an empirical-Bernstein bound that uses the sample variance $\hat{V}_m(\lambda)$, which is much tighter when the calibration losses are low-variance (which they are here, since most routed items are correct). Betting-based confidence sequences also work if you reuse the calibration stream while expanding the price set. We kept Hoeffding in the main results for simplicity and comparability with PAC.

Q5. Why no union bound over λ — isn’t that cheating?

No, it's the equal-cost structure. As λ grows the threshold $1/\lambda$ only falls, so an item can only switch AI→expert, never between cheap sources. So $L_T(\lambda)$ is monotone and the candidate policies form one nested chain; the bad-selection event collapses onto a single fixed crossing price, so one pointwise Hoeffding bound suffices. The moment costs are unequal, source selection depends on λ , monotonicity breaks, and you *do* pay a union bound over a price grid (or over the $\leq 1+(T+m)\binom{k+1}{2}$ breakpoints).

Q6. Why does the online error sometimes exceed ε ?

The guarantee is $\frac{1}{T} \sum_t \ell(Y_t, \tilde{Y}_t) \leq \varepsilon + B_T$, with $B_T = \Lambda_{\text{on}}/(\eta T) + \frac{1}{p_{\min}} \sqrt{2 \log(1/\delta)/T}$. The price update centers on ε *directly*, so the bound sits exactly at $\varepsilon + B_T$, not at ε : there is a finite-horizon cushion B_T that the target is allowed to overshoot by. At small ε and small p (few audits, larger B_T) finite- T noise can push the realized error a little above ε — e.g. ImageNetV2, $\varepsilon = 0.05$, $p = 0.1$. Since $B_T = O(T^{-1/2})$, the cushion shrinks as the stream lengthens or the audit rate rises, and the error settles back to ε . (An earlier draft carried a separate internal target ε_0 and “spent” this slack explicitly; the current version folds it into B_T .)

Q7. How do you choose the audit rate p (and the step size η)?

p is a deliberate cost/validity dial: small p saves budget but gives noisier feedback (larger $1/p_{\min}$ term); large p corrects more labels and lowers realized error. The floor $p_{\min}$ enters the bound as $1/p_{\min}$, so you don't want it too small. For η , the bound is $\Lambda_{\text{on}}/(\eta T) + O(1/\sqrt{T})$, optimized at $\eta \asymp T^{-1/2}$, which makes the whole slack $O(T^{-1/2})$. In the runs we used a fixed $\eta = 1$ and still saw control because T is large.

Q8. If the dashed counterfactual AI-error curve already satisfies the error bound, are audits actually needed?

The dashed curve is not “use AI on every item.” It is the error of the *selected* AI labels after the deferral rule has already removed high-proxy-loss items: $H_t = (1 - D_t)\ell(Y_t, \hat{Y}_t)$. So the fact that it is below ε says the confidence proxy and threshold/router are working well in this dataset. It does *not* mean the online learner could certify or adapt without labels. Audits provide the feedback for updating λ_t and protect against miscalibration or stream shift; without audits, AI-label losses are hidden and a bad threshold could silently violate the target. The right takeaway is that a strong proxy may reduce the number of audits needed, not that audits are unnecessary.

Q9. Is this regret or error control? How does it relate to online conformal prediction?

It's *realized-error* control over a fixed horizon, not regret against an offline expert-query oracle. The lineage is online/adaptive conformal prediction with partial (semi-bandit) feedback — those methods update a calibration parameter online, and the key borrowed idea is that when labels aren't always revealed you must correct the update for the observation process. Our inverse-probability signal O_t/q_t is exactly that correction. The object we control is the average error of the *released dataset*, not prediction-set coverage.

Q10. What's the actual cost of calibration / auditing — isn't that hidden expert spend?

Good — we separate three costs in the report. *Deployment* cost is $C_T(\lambda)$, the quantity we minimize. *Development* cost is the expert labels used to train the proxy and to certify the price ($m = 0.2T$ here); if those labeled items are part of the pool and corrected in the release, there’s no duplicate expert charge for them. *Audit* cost online is $\mathbb{E}[C_t \mid \mathcal{G}_t] = D_t + (1 - D_t)p_t$ per round — the extra expert spend to obtain feedback. So the headline “78% saved” is deployment savings; calibration is a one-time, accountable overhead.

Q11. Does the guarantee need i.i.d. data?

The offline pool guarantee is *conditional on the fixed pool* — it only needs the calibration indices to be sampled i.i.d. from the pool, not the data to be i.i.d. The *generalization* corollary (population risk + fresh deployment pool) does assume an i.i.d. source distribution. The online guarantee is fully *distribution-free in the data sequence* — it can even be adversarial — with all randomness coming from the audit coin flips. That robustness is one of the nicer features of the online side.

Q12. Why is ImageNetV2 harder (lower savings, occasional violation)?

ImageNetV2 is a known distribution shift from ImageNet — the base model is less accurate and less calibrated on it, so at the same ε fewer items clear the confidence threshold and we defer more (savings drop from ~ 0.78 to ~ 0.55 at $\varepsilon = 0.10$). It’s a useful stress test precisely because the proxy is worse there, and the guarantee still holds — consistent with “proxies affect cost, not validity.”